

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
K.D. PATEL DEPARTMENT OF INFORMATION TECHNOLOGY

ACADEMIC YEAR: 2024-25

List of Experiments

Subject: Ethical Hacking (OCIT3001) (6th Semester)

Sr. No.	Name of Experiment	Hours
1.	A Virtual Private Cloud (VPC) network is a private cloud hosted within a public cloud, enabling organizations to use the public cloud's resources while being completely isolated from other cloud users. A VPC provides networking functionality to Compute Engine virtual machine (VM) instances, Kubernetes Engine containers, and other Google Cloud services. Each Google Cloud project by default has a default VPC network, which provides each region with an automatically-created subnet network. In this experiment, you'll learn how to use Cloud Shell to create a custom VPC network with subnets. Create a VPC using Cloud Shell. Reference: https://www.cloudskillsboost.google/paths/419/	04
2.	Reports are essential for helping remediate findings, especially for cybersecurity, compliance, and quality assurance. These reports often highlight vulnerabilities, issues, or non-compliance with established standards. By analyzing these reports, you can identify areas for improvement and gather data-driven evidence to make informed decisions. Concrete data enables efficient prioritization and resource allocation to address the identified issues. Consequently, addressing the findings outlined in reports helps mitigate potential risks and vulnerabilities that could otherwise be exploited. This is especially critical in cybersecurity, where unaddressed issues can lead to data breaches or system compromises. As a security analyst, you are responsible for evaluating controls against established standards to ensure that an organization's security posture is effective, compliant, and aligned with industry best practices. This evaluation process is crucial for helping with risk management, compliance, and continuous security improvement, ultimately helping organizations protect sensitive data, systems, and their overall reputation. In this experiment, you'll use the Security Command Center interface to identify and remediate threats and vulnerabilities and confirm that the issues have been resolved. Reference: https://www.cloudskillsboost.google/paths/419/	02

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3.	IAM, or Identity and Access Management, is a collection of processes and technologies that help organizations manage digital identities in their environment. With IAM, access control is managed by defining the identity of users and their roles in relation to available resources. Resource access permissions are not granted directly to individual users. Instead, users are assigned to roles that are then given to authenticated principals. While the term "members" was used in the past, IAM now refers to these individuals as principals, although some APIs still use the previous terminology. There are three types of IAM roles in Google Cloud: Basic roles: Roles historically available in the Google Cloud console. These roles are Owner, Editor, and Viewer. Predefined roles: Roles that give finer-grained access control than the basic roles. For example, the predefined role Pub/Sub Publisher (roles/pubsub.publisher) provides access to only publish messages to a Pub/Sub topic. Custom roles: Roles that you create to tailor permissions to the needs of your organization when predefined roles don't meet your needs. In this experiment, you'll learn how to create and manage Identity and Access Management (IAM) custom roles. Reference: https://www.cloudskillsboost.google/paths/419/	02
4.	The assets within a cloud environment need to be protected from unauthorized access. To address this, security professionals use perimeter protection which refers to the security measures implemented to defend the edge of a network or system against unauthorized access and cyber threats. One type of perimeter protection includes using firewalls to manage and secure network traffic entering and leaving a cloud environment. Firewalls help protect trusted internal networks (like a company's private network) from untrusted external networks (such as the Internet). Firewalls examine both incoming and outgoing network traffic based on predefined rules to either allow or block specific data packets. This is crucial for helping maintain application security, traffic control, compliance, and policy enforcement. In this experiment, you'll access a firewall and create rules to test the security of a server and make modifications as necessary. Reference: https://www.cloudskillsboost.google/paths/419/	02
5.	Identifying vulnerabilities and implementing remediation techniques is crucial for helping ensure the security and stability of various systems and applications. Many applications and systems handle sensitive information, such as personally identifiable information, financial records, or intellectual property. Identifying vulnerabilities helps	02

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	protect this sensitive data from unauthorized access and potential breaches. Addressing vulnerabilities early in the development process is generally more cost-effective than dealing with security breaches later. The cost of remediating a vulnerability is often much higher than the cost of preventing it in the first place. As a security analyst, regularly scanning for vulnerabilities can help identify and address weaknesses before malicious attacks, thus mitigating potential threats proactively. It provides insight into an application's attack surface, helping enable you to understand potential avenues of exploitation and prioritize critical areas for improvement. In this experiment, you'll not only learn how to set up and run a vulnerable application but scan it for vulnerabilities. Reference: https://www.cloudskillsboost.google/paths/419/	
6.	Firewall rules play a crucial role in cloud network security because they control which traffic is allowed to enter and leave your cloud environment. The nature of cloud networking is ever changing and complex. Organizations' needs change, new cloud resources are added or removed, and new vulnerabilities are discovered. This means that you'll need to continually adjust and maintain the configuration of firewall rules to adapt with these changes. Tracking and managing critical updates in your cloud environment can be challenging. Luckily, you can manage the configuration of your cloud resources by leveraging infrastructure as code (IaC). IaC is the provisioning and managing of infrastructure through using reusable scripts. It can be used to automate workflows such as updating firewall rules. Terraform and Cloud Shell make it easy to manage and update firewall rules efficiently. By writing desired firewall rules in Terraform configuration files and executing Terraform commands in Cloud Shell, you can ensure that your network security policies remain consistent and version-controlled. This approach will help you maintain and track changes over time. In this experiment, you'll learn how to clone a Terraform repository and deploy a VPC network and firewall. Reference: https://www.cloudskillsboost.google/paths/419/	04
7.	Encryption is a critical component for protecting data in cloud environments. Security professionals use cryptography to transform information into a form that unintended readers can't understand. Symmetric and asymmetric keys are cryptographic tools used to secure data and enable secure communication over networks. Each type of key has its own distinct differences and can be deployed for different situations. Symmetric Keys:	02

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	Symmetric key cryptography uses a single key to encrypt and decrypt data. The same key is used by both the sender and the recipient which is why it's called symmetric. Symmetric key cryptography is efficient and fast. Asymmetric Keys: Asymmetric key cryptography (also known as public-key cryptography) uses a pair of keys: a public key and a private key. One of the keys is used to encrypt data, while the other key decrypts data. These keys are mathematically related but cannot be derived from each other. Asymmetric cryptography is known for its slow performance. This is due to the use of these two mathematically related keys which are longer than those used in symmetric encryption. In practice, many secure communication systems use a combination of symmetric and asymmetric cryptography to achieve both efficiency and security. For example, the Hypertext Transfer Protocol Secure (HTTPS) protocol uses asymmetric cryptography for the initial handshake to establish a secure connection, and then switches to symmetric encryption for the actual data transfer. One of the major problems with symmetric key cryptography involves key distribution. How do you ensure the secure exchange of keys without having the key be compromised or stolen? Asymmetric key cryptography solves this problem by using a public and private key pair. However, it is computationally more expensive, so it's commonly used for initial key exchange and digital signatures, while symmetric keys are used for the bulk encryption of data. In this experiment, you'll create both a symmetric key and an asymmetric key to address a request for more space to securely store data. Reference: https://www.cloudskillsboost.google/paths/419/	
8.	Event Threat Detection is one of Security Command Center's (SCC) services. Event Threat Detection is a log-based threat analysis that continuously monitors Google Cloud logs for potential threats. When Event Threat Detection identifies suspicious activity, it generates a finding that you can investigate. In this experiment, you'll analyze findings in the Google Cloud Security Command Center and examine related events in Cloud Logging. Reference: https://www.cloudskillsboost.google/paths/419/	02
9.	A false positive is an alert that incorrectly detects the presence of a threat. False positives can be triggered by genuine and legitimate user activity. Security teams may spend lots of time and resources investigating false positive alerts only to discover that there is no	02

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	real threat. In this experiment, you'll recreate the activity that generates a false positive alert. Then, you'll access and analyze the false positive threat using Security Command Center (SCC) and take action to address it. You'll be using two separate accounts in this experiment: one account to trigger the false positive and another account to analyze and remediate the false positive. Reference: https://www.cloudskillsboost.google/paths/419/	
10.	Google Cloud services write audit logs that record administrative activities and access within your Google Cloud resources. Audit log entries help you answer the questions "who did what, where, and when" within your Google Cloud projects. Enabling audit logs helps your security, auditing, and compliance entities monitor Google Cloud data and systems for possible vulnerabilities or external data misuse. In this experiment, you'll investigate audit logs to identify patterns of suspicious activity involving cloud resources. Reference: https://www.cloudskillsboost.google/paths/419/	02
11.	Business continuity and disaster recovery planning is critical for sustaining business operations while recovering from a significant security incident, natural disaster, or disruption. Google Cloud Backup and DR Service is a cloud-based backup and disaster recovery solution that enables the backup and recovery of data, to support quick resumption of critical business operations. After Backup and DR performs an initial full backup, your data (general applications, VMware VMs, Compute Engine VMs, databases, and file systems) is backed up incrementally, updating and storing any data that has changed since the last backup. The initial configuration of the Backup and DR service includes the deployment of a management appliance that can take up to 45 minutes to complete. This task has been carried out for you prior to the lab startup. Once the Backup and DR Service are enabled, you can explore the Backup and DR management console and protect workloads. This experiment guides you through the steps of discovering and protecting a Compute Engine instance, and finally mounting a fully-functional new Compute Engine instance from the backup image to a new location. Reference: https://www.cloudskillsboost.google/paths/419/	02
12.	In this experiment, you'll apply your knowledge of cloud cybersecurity to identify and remediate vulnerabilities. You'll be given a scenario, and a set of tasks to complete in	04

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	Google Cloud Security Command Center. These tasks will require you to use your skills to work to analyze and remediate active vulnerabilities relating to a security incident, answer questions about the vulnerabilities, and complete challenges that will assess your cloud cybersecurity skills. There are also a number of challenges in the experiment. A challenge is a task where you will be asked to complete the task on your own without instructions. By successfully completing this experiment, you will demonstrate your ability to identify, prioritize, and remediate security vulnerabilities and misconfigurations within the cloud environment. These are essential skills to enhance the security posture of Google Cloud environments, reducing the risk of data breaches, unauthorized access, and other security incidents. Reference: https://www.cloudskillsboost.google/paths/419/	
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